

Grade 11 - Term 3

C1–C4 Practice Exercises and Solutions

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Evaluated criteria

- C1** Characterises continuous random variables by distinguishing them from discrete variables and interpreting their probability density function, cumulative distribution function, and support in economic contexts.
- C2** Applies uniform and triangular continuous distributions with constant or piecewise-linear probability density functions to calculate interval probabilities using geometric areas.
- C3** Applies circular-arc probability density functions to calculate interval probabilities using the geometry of circles and circular segments.
- C4** Applies the exponential distribution to constant-rate waiting-time situations and assesses its suitability.

1 Identifying Probability Distributions from Contexts [C1-5]

A variety of Colombian organizations use probability models to represent uncertain quantities and events. In each situation below, a random variable is defined from the information provided. The appropriate probability distribution must be identified from the characteristics of the situation.

Questions/Tasks.

- C1** For each context, characterize the random variable and identify the appropriate probability distribution. Briefly justify your choice based on the features described. [5]

Context 1. A payment platform monitors one scheduled electronic transfer. By the end of processing, it is either completed successfully or not completed. The variable X_1 records the outcome to help estimate verification and support needs.

Context 2. An agricultural technology company monitors the daily temperature variation inside a refrigerated storage unit. The variation X_2 ranges from -2°C to 2°C , with all values in this interval considered equally likely. The company uses this model to set an operating margin.

Context 3. A streaming platform sends the same offer to exactly 80 randomly selected eligible users. Each user either subscribes or does not. The variable X_3 counts the number of subscribers to estimate the server capacity needed for the promotion.

Context 4. A Bogotá parking company records the number of vehicle breakdowns during a fixed one-week period. Comparable records show that breakdowns occur independently and without a predictable timetable. The variable X_4 counts the breakdowns to help schedule maintenance staff.

Context 5. An online marketplace studies the delivery time X_5 for a particular type of order. Delivery times range from 20 to 80 minutes, with deliveries around 45 minutes occurring most frequently. They become progressively less common toward either extreme. The company uses this model to estimate delivery times.

Context 6. A quality-control laboratory tests electronic components until the first defective component is found. Components are tested independently, and testing stops after the first defect. The variable X_6 records the number of components tested, including the defective one, to estimate inspection costs.

Context 7. A cloud-computing provider studies storage usage X_7 among small business customers. Usage ranges from 0 to 50 gigabytes and becomes steadily more common toward the upper limit, without reaching a peak before 50 GB. The provider uses this model to plan capacity.

Context 8. A retailer sends a promotional message to exactly 120 customers selected using the same procedure. Each customer either makes a purchase or does not. The variable X_8 counts the number of purchases to estimate the inventory needed for the campaign.

Context 9. A coffee exporter measures the moisture content X_9 of coffee bags under usual storage conditions. Measurements cluster around 11

Context 10. A food-delivery company records the number of customer cancellations during a fixed three-hour dinner period in one delivery zone. Comparable evenings show cancellations occurring independently without a recurring timetable. The variable X_{10} counts the cancellations to estimate customer-service needs.

C1 - Distribution identification: Solutions

[10]

The appropriate distribution for each context is determined by whether the variable is discrete or continuous and by the defining features of the situation:

1. **Context 1 — Discrete, Bernoulli.** The transfer is either completed or not completed.
2. **Context 2 — Continuous, Uniform.** All values within a fixed interval are equally likely.
3. **Context 3 — Discrete, Binomial.** The variable counts subscriptions among 80 comparable users.
4. **Context 4 — Discrete, Poisson.** The variable counts independent incidents occurring during a fixed period.
5. **Context 5 — Continuous, Triangular.** The variable has a minimum, maximum, and most likely value.
6. **Context 6 — Discrete, Geometric.** The variable counts trials until the first defective component is found.
7. **Context 7 — Continuous, Linear.** The density increases steadily toward the upper limit.
8. **Context 8 — Discrete, Binomial.** The variable counts purchases among 120 comparable customers.
9. **Context 9 — Continuous, Normal.** Values cluster symmetrically around a central mean.
10. **Context 10 — Discrete, Poisson.** The variable counts independent cancellations during a fixed



2 Delivery Time for a Local Grocery Order

[C2-4]

A Colombian grocery-delivery company studies the additional time required to deliver orders during a busy evening period. For a particular delivery zone, the added delivery time X can range from 0 to 10 minutes. A delay of any

length within this interval is possible, and shorter delays become steadily more common. Historical data indicate that no additional delay is 4 times as common as a 10-minute delay.

Questions/Tasks.

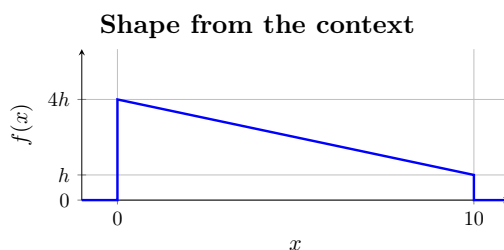
C2 Determine the complete probability density function for the added delivery time. Then determine the probability that the added delivery time exceeds 7 minutes, and the probability that it is between 3 and 7 minutes. [4]

C2 - Linear probability distribution

[4]

2.1 Support and Shape

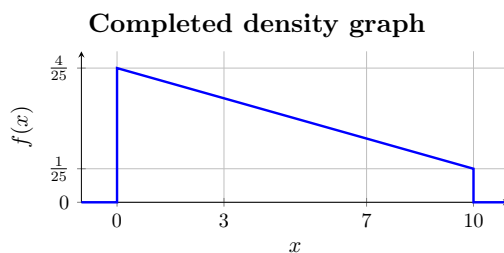
The random variable X has support $[0, 10]$. The context describes a steadily decreasing straight-line density whose height at 0 is four times its height at 10. Write the endpoint heights as $4h$ and h . The density is zero outside the support.



2.2 Total Area = 1

The total area under a probability density function must equal 1. Therefore, the trapezoid formed by the density has area 1:

$$1 = \frac{1}{2}(10)(4h + h) = 25h, \quad h = \frac{1}{25}, \quad 4h = \frac{4}{25}.$$



2.3 Straight Line Rule

The density is a straight line passing through the two endpoint points

$$(x_1, y_1) \quad \text{and} \quad (x_2, y_2). \quad \longrightarrow \quad m = \frac{y_2 - y_1}{x_2 - x_1}.$$

$$\left(0, \frac{4}{25}\right) \quad \text{and} \quad \left(10, \frac{1}{25}\right). \quad \longrightarrow \quad m = \frac{\frac{1}{25} - \frac{4}{25}}{10 - 0} = -\frac{3}{250},$$

and the y -intercept is

$$\begin{aligned} (x_1, y_1) \quad \text{and} \quad (x_2, y_2) . \quad \longrightarrow \quad b = y_1 - mx_1 = y_2 - mx_2. \\ \left(0, \frac{4}{25}\right) \quad \text{and} \quad \left(10, \frac{1}{25}\right) . \quad \longrightarrow \quad b = \frac{4}{25} - \left(-\frac{3}{250}\right)(0) = \frac{1}{25} - \left(-\frac{3}{250}\right)(10) = \frac{4}{25}. \end{aligned}$$

Therefore, the probability density function is

$$f_X(x) = \frac{40 - 3x}{250}$$

on the stated interval.

2.4 Complete pdf

Including the values outside the possible range of added delivery times, the complete probability density function is

$$f_X(x) = \begin{cases} \frac{40 - 3x}{250}, & 0 \leq x \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

2.5 Probability of More Than 7 Minutes

The company wants to know the chance that an order takes more than 7 additional minutes to deliver. This corresponds to the area under the density from 7 to the maximum possible delay of 10 minutes:

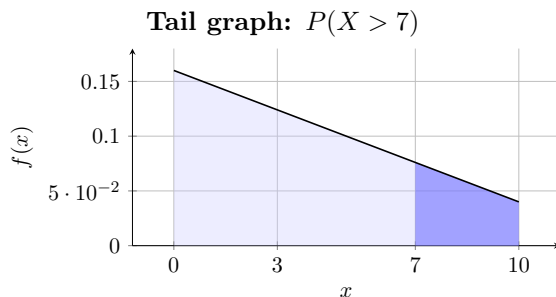
$$P(X > 7) = \frac{1}{2}(10 - 7)(f_X(7) + f_X(10)).$$

Since

$$\begin{aligned} f_X(7) &= \frac{19}{250} \quad \text{and} \quad f_X(10) = \frac{1}{25}, \\ P(X > 7) &= \frac{1}{2}(3) \left(\frac{19}{250} + \frac{1}{25} \right) = \frac{9}{100}. \end{aligned}$$

Thus, the probability of more than 7 minutes of additional delivery time is

$$P(X > 7) = 0.09 .$$



2.6 Probability Between 3 and 7 Minutes

The company also wants to know the chance that the added delivery time falls between 3 and 7 minutes. This corresponds to the area under the density between these two times:

$$P(3 \leq X \leq 7) = \frac{1}{2}(7 - 3)(f_X(3) + f_X(7)).$$

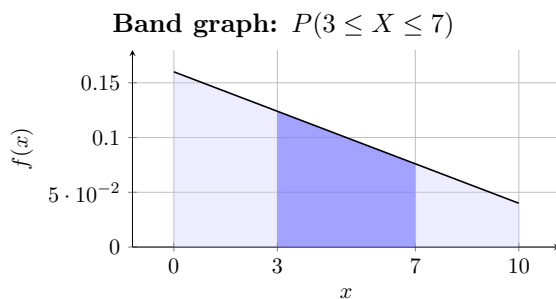
Since

$$f_X(3) = \frac{31}{250} \quad \text{and} \quad f_X(7) = \frac{19}{250},$$

$$P(3 \leq X \leq 7) = \frac{1}{2}(4) \left(\frac{31}{250} + \frac{19}{250} \right) = \frac{2}{5}.$$

Thus, the probability of between 3 and 7 minutes of additional delivery time is

$$P(3 \leq X \leq 7) = 0.40.$$



3 Daily Commute Time for a Bogotá Apprentice

[C2-4]

An apprentice records the time spent travelling to a training programme each day. Commute times range from 1 to 9 hours. Times become steadily more common from 1 hour up to a most common time of 4 hours, then steadily less common through 9 hours; there are no commute times right at either limit.

Questions/Tasks.

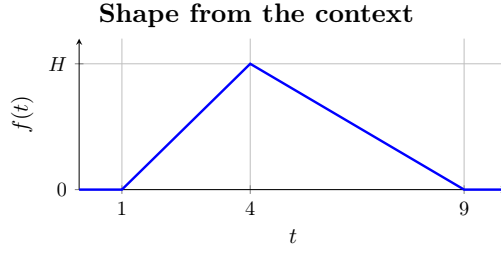
C2 Determine the complete probability density function for the commute time. Then determine the probability that the commute takes more than 6 hours, and the probability that it takes between 3 and 7 hours. [4]

C2 - Triangular probability distribution

[4]

3.1 Support and Shape

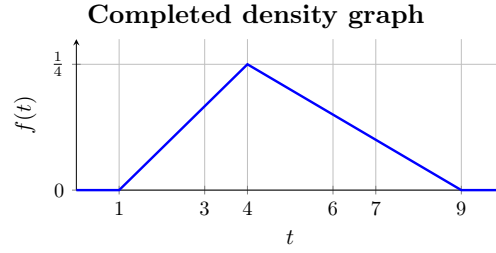
The random variable T has support $[1, 9]$. The context supplies zero height at both endpoints, a steady rise to the most common value at $t = 4$, and a steady fall thereafter. Denote the unknown peak height by H ; the density is zero outside the support.



3.2 Total Area = 1

The triangle must have area 1:

$$1 = \frac{1}{2}(9 - 1)H = 4H, \quad H = \frac{1}{4}.$$



3.3 Straight Line Rule

For the rising segment, the density is a straight line passing through the two endpoint points

$$(x_1, y_1) \quad \text{and} \quad (x_2, y_2). \quad \longrightarrow \quad m = \frac{y_2 - y_1}{x_2 - x_1}.$$

$$(1, 0) \quad \text{and} \quad \left(4, \frac{1}{4}\right). \quad \longrightarrow \quad m_1 = \frac{\frac{1}{4} - 0}{4 - 1} = \frac{1}{12}.$$

The y -intercept is

$$(x_1, y_1) \quad \text{and} \quad (x_2, y_2). \quad \longrightarrow \quad b = y_1 - mx_1 = y_2 - mx_2.$$

$$(1, 0) \quad \text{and} \quad \left(4, \frac{1}{4}\right). \quad \longrightarrow \quad b_1 = 0 - \frac{1}{12}(1) = -\frac{1}{12}.$$

Therefore, the equation of the rising segment is

$$y = m_1 t + b_1 = \frac{t - 1}{12}.$$

For the falling segment, the density is a straight line passing through the two endpoint points

$$(x_1, y_1) \quad \text{and} \quad (x_2, y_2). \quad \longrightarrow \quad m = \frac{y_2 - y_1}{x_2 - x_1}.$$

$$\left(4, \frac{1}{4}\right) \quad \text{and} \quad (9, 0). \quad \longrightarrow \quad m_2 = \frac{0 - \frac{1}{4}}{9 - 4} = -\frac{1}{20}.$$

The y -intercept is

$$(x_1, y_1) \quad \text{and} \quad (x_2, y_2). \quad \longrightarrow \quad b = y_1 - mx_1 = y_2 - mx_2.$$

$$\left(4, \frac{1}{4}\right) \quad \text{and} \quad (9, 0). \quad \longrightarrow \quad b_2 = \frac{1}{4} - \left(-\frac{1}{20}\right)(4) = 0 - \left(-\frac{1}{20}\right)(9) = \frac{9}{20}.$$

Therefore, the equation of the falling segment is

$$y = m_2t + b_2 = \frac{9 - t}{20}.$$

The adjacent rules agree at $t = 4$: both give $\frac{1}{4}$.

3.4 Complete pdf

Including the values outside the possible range of commute times, the complete probability density function is

$$f_T(t) = \begin{cases} \frac{t-1}{12}, & 1 \leq t \leq 4, \\ \frac{9-t}{20}, & 4 < t \leq 9, \\ 0, & \text{otherwise.} \end{cases}$$

3.5 Probability of More Than 6 Hours

The apprentice wants to know the chance that the commute takes more than 6 hours. This corresponds to the area under the density from 6 to the maximum possible commute time of 9 hours:

$$P(T > 6) = \frac{1}{2}(9 - 6)(f_T(6) + f_T(9)).$$

First, evaluate the density at the two endpoints:

$$f_T(6) = \frac{9 - 6}{20} = \frac{3}{20},$$

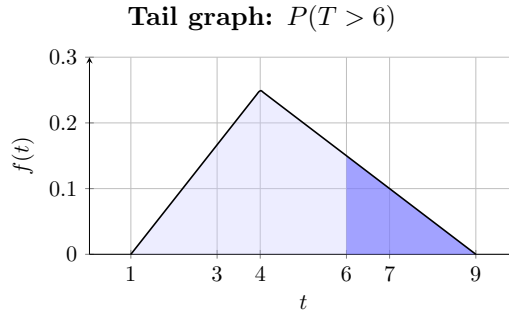
$$f_T(9) = \frac{9 - 9}{20} = 0.$$

Substituting these values gives

$$\begin{aligned} P(T > 6) &= \frac{1}{2}(9 - 6) \left(\frac{3}{20} + 0 \right) \\ &= \frac{1}{2}(3) \left(\frac{3}{20} \right) \\ &= \frac{9}{40}. \end{aligned}$$

Thus, the probability of a commute lasting more than 6 hours is

$$P(T > 6) = 0.225.$$



3.6 Probability Between 3 and 7 Hours

The apprentice also wants to know the chance that the commute takes between 3 and 7 hours. We use the complement rule: the event $3 \leq T \leq 7$ is the complement of the two tail events $T < 3$ and $T > 7$. Therefore,

$$P(3 \leq T \leq 7) = 1 - P(T < 3) - P(T > 7).$$

The two complementary regions are triangular, so their probabilities are given by

$$P(T < 3) = \frac{1}{2}(3-1)f_T(3), \quad P(T > 7) = \frac{1}{2}(9-7)f_T(7).$$

Hence,

$$P(3 \leq T \leq 7) = 1 - \frac{1}{2}(3-1)f_T(3) - \frac{1}{2}(9-7)f_T(7).$$

First, evaluate the density at the two relevant points:

$$f_T(3) = \frac{3-1}{12} = \frac{1}{6},$$

$$f_T(7) = \frac{9-7}{20} = \frac{1}{10}.$$

Substituting these values gives

$$\begin{aligned} P(3 \leq T \leq 7) &= 1 - \frac{1}{2}(3-1) \left(\frac{1}{6} \right) - \frac{1}{2}(9-7) \left(\frac{1}{10} \right) \\ &= 1 - \frac{1}{6} - \frac{1}{10} \\ &= \frac{11}{15}. \end{aligned}$$

Thus, the probability of a commute lasting between 3 and 7 hours is

$$P(3 \leq T \leq 7) = \frac{11}{15} \approx 73.3\% .$$

